The target-free clique conjecture for threshold-linear networks: complete exact maps for $n \leq 7$, infinite families, and unbounded thresholds

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Abstract

For combinatorial threshold-linear networks (CTLNs), Curto and Morrison asked whether the supports of stable fixed points are exactly the target-free cliques of the underlying digraph. Geneson disproved this with a six-node graph near $q = 1$, where $q = \delta(1-\varepsilon)/\varepsilon > 1$ is the legality parameter, and proved that the conjecture holds once $q \geq n - 2 - (n-3)\varepsilon/2$. We study the gap between these two results with exact arithmetic throughout. First, a reduction theorem shows that at every legal parameter the conjecture on n nodes is equivalent to the statement that no non-complete digraph on $k \leq n$ nodes has a positive, linearly stable full-support fixed point. Second, using the rank-one structure of the weight matrix, an exact $\varepsilon \to 0$ spectral criterion, Hurwitz determinants with a three-valued treatment of boundary points, a cylindrical decomposition of the two-parameter legal region, and a ray reduction to the legality curve, we obtain complete certified maps of stable non-clique full supports: there are none for $n \leq 5$; at $n = 6$ exactly three of the 1,540,944 digraphs have one; at $n = 7$ exactly 124 of the 882,033,440 digraphs do, one of which is invisible in the $\varepsilon \to 0$ limit. The exact thresholds are $q_6^* \approx 1.1664$, a root of an explicit sextic, and $q_7^* = \rho \approx 1.3247$, the plastic number: for $n \leq 7$ the conjecture holds at every legal parameter with $q \geq \rho$ and fails for every $q \in (1, \rho)$. Third, closed-twin blow-ups yield infinite families: counterexamples on every $n \geq 6$ nodes, finite-ε-only counterexamples on every $n \geq 7$ nodes, a proof that the critical value q_n^* tends to infinity, and a single legal parameter point at which stable non-clique supports of every size $n \geq 6$ exist. Several of the necessary conditions used to prune the search are machine-checked in Lean 4.

1 Introduction

1.1 The conjecture

A *threshold-linear network* on n neurons is the dynamical system

$$\frac{dx_i}{dt} = -x_i + \Big[\sum_{j=1}^{n} W_{ij} x_j + \theta\Big]_+, \qquad i = 1, \ldots, n, \tag{1}$$

with $[u]_+ = \max(u, 0)$ and a constant external drive $\theta > 0$. A *combinatorial* TLN (CTLN) builds W from a directed graph G and two parameters $\varepsilon, \delta > 0$: weak inhibition $-1 + \varepsilon$ along edges and strong inhibition $-1 - \delta$ along non-edges (Section 2). The CTLN is *legal* when $\delta > 0$, $0 < \varepsilon < \delta/(\delta + 1)$, which we write as $q := \delta(1 - \varepsilon)/\varepsilon > 1$. Fixed points of (1) are indexed by their supports $\sigma \subseteq [n]$, and a central theme of the CTLN programme is to read the fixed-point structure off the graph [3].

A subset σ is a *clique* if every pair of distinct nodes of σ is joined by edges in both directions; a node $k \notin \sigma$ is a *target* of σ if every node of σ sends an edge to k; σ is *target-free* if it is a clique with no target. Open Question 10 of Curto and Morrison [3] is the *target-free clique conjecture*: *the supports*

of stable fixed points of a CTLN are exactly its target-free cliques. One direction is a classical graph rule: every target-free clique supports a stable fixed point at every legal parameter (Lemma 2.5). The content of the conjecture is the converse.

1.2 Geneson's counterexample and bound

Geneson [4] disproved the conjecture with an explicit graph on six nodes and $\delta = \frac{29}{25}\varepsilon$ for all sufficiently small ε (so $q \approx 29/25$), and proved that it does hold whenever

$$q \;\geq\; n - 2 - \tfrac{(n-3)\varepsilon}{2}. \tag{2}$$

This leaves a gap between q slightly above 1 and q near $n-2$. The present paper maps that gap exactly for $n \leq 7$ and studies its behaviour for large n.

1.3 Our results

Throughout, results are labelled PROVED (a proof is given here or in the project notes), CERTIFIED (an exact rational or algebraic computation recorded in a re-runnable certificate file), or LEAN (machine-checked in Lean 4 with Mathlib; marked ^{L} in statements). No floating-point number enters any decision; decimals in the text are display approximations of exactly specified algebraic numbers. Graphs on $n \leq 7$ nodes reported as counterexamples have been checked by hand (the two new six-node graphs also by independent adversarial verification workflows, the 124 seven-node graphs also by four independently written exact verifiers); family members on $n \geq 8$ nodes are certified instances of proved theorems but have *not* been hand-verified (Section 7).

Reduction and exact machinery. At every legal parameter, the conjecture on n nodes is equivalent to the absence of a non-complete digraph on $k \leq n$ nodes whose full support is a positive, linearly stable fixed point (Theorem 2.6). The full-support problem has a rank-one structure: positivity depends on one parameter only, and the $\varepsilon \to 0$ limit is an exact spectral condition on the adjacency matrix compressed to $\mathbf{1}^{\perp}$ (Lemmas 2.2 and 2.4). Stability at some legal point is equivalent to stability on the one-dimensional legality curve (Lemma 3.4). Together with five necessary conditions valid at every parameter (domination, positivity, an edge bound, Stodola's condition and a twin obstruction, Section 3.4) these reduce the problem to a manageable set of exact one- and two-parameter certifications.

Complete maps for $n \leq 6$. No digraph on at most five nodes is a counterexample at any legal parameter. Among the 1,540,944 digraphs on six nodes exactly three non-complete graphs have a stable positive full support at some legal parameter: Geneson's graph, a partner obtained from it by a bidirected in-twin switch, and one further 18-edge graph (Theorem 4.2). The critical value is $q_6^* \approx 1.1664$, the root in $(561/481, 7/6)$ of $797q^6 - 1686q^5 + 2159q^4 - 3004q^3 + 2499q^2 - 2158q + 1521$ (Theorem 4.4).

The complete map at $n = 7$. Every one of the 882,033,440 digraphs on seven nodes is either excluded by a proved necessary condition or carries an exact two-parameter certificate. Exactly 124 non-complete graphs have a stable positive full support at some legal parameter (Theorem 5.1). Of these, 123 are visible in the $\varepsilon \to 0$ limit and one is not: its compressed adjacency has spectral abscissa exactly $-1/2$ and it is stable only in a window of moderate ε. The natural monotonicity statement "stable at finite ε implies stable in the limit" is false (Proposition 5.3), and an exact first-order coefficient R^* decides when a stable region bulges past its limit boundary (Theorem 5.4). The threshold is $q_7^* = \rho$, the plastic number (real root of $q^3 = q+1$), and for $n \leq 7$ the conjecture holds at every legal parameter with $q \geq \rho$ and fails for every $q \in (1, \rho)$ (Theorem 5.6).

Infinite families and unbounded thresholds. Closed-twin blow-ups preserve full-support fixed-point data in closed form (Lemmas 6.1 and 6.2). This yields: counterexamples on every $n \geq 6$ (Theorem 6.3); finite-ε-only counterexamples on every $n \geq 7$ (Theorem 6.5); families whose thresholds approach 1.84, 3 and 3.96 (Theorem 6.6); and, via an in-tree construction realising prescribed limit polynomials, a proof that $q_n^* \to \infty$ (Theorem 6.7). Finally, at the single legal point $(\varepsilon, \delta) = (1/21, 11/210)$ there are stable non-clique full supports on n nodes for every $n \geq 6$, and on no $n \leq 5$ (Theorem 6.8).

Verification. Every count and critical value is backed by a re-runnable certificate file (Appendix A); the pipelines and theorems were independently audited; all $n \leq 7$ counterexamples were hand-verified, and the 124 seven-node ones were re-decided by four independently written exact verifiers; and 87 supporting theorems are machine-checked (Section 7).

1.4 Related work

The conjecture and the surrounding graph rules are from Curto and Morrison [3]. Graphical domination, which we use as a prune in its verbatim form, is due to Curto [2]. Geneson [4] gave the first counterexample and the bound (2); he does not pose or treat the gap question. An unrefereed preprint, "Six Is Minimal" [1], obtained independently and by different exact methods that no digraph on at most five nodes has a stable non-clique support, used Geneson's graph as the six-node witness, computed the full-support region of that graph, and gave a family on $5r+6$ nodes. Their conventions agree with ours up to a transpose of the non-edge matrix; their exclusion for $n \leq 5$ agrees with our $n = 5$ map; their region for Geneson's graph gives $1 + u^* = 1.16635502785\ldots$, the same sextic root as our q_6^*; and their family is a clique-class blow-up of the same seed with $q \to 1$, an instance of Lemma 6.1. Their work does not classify $n = 6$ or treat $n \geq 7$, finite-ε-only examples, the reduction theorem or growing thresholds; it does give an explicit rational parameter schedule for its family and global nondegeneracy certificates, which prompted our nondegeneracy check. We regard the two works as independent and consistent. Our comparison is based on reading their repository; we did not run their code.

2 Preliminaries

2.1 Conventions

A digraph G on $[n]$ is encoded by its adjacency matrix A, with

$$A_{ij} = 1 \iff \text{there is an edge } i \to j \qquad \text{(row } i \text{ lists the nodes that } i \text{ sends to).}$$

The CTLN weight matrix is

$$W_{ij} = \begin{cases} 0, & i = j, \\ -1+\varepsilon, & i \neq j,\ A_{ji} = 1 \quad (j \to i:\ \text{weak inhibition}), \\ -1-\delta, & i \neq j,\ A_{ji} = 0 \quad (j \not\to i:\ \text{strong inhibition}). \end{cases} \tag{3}$$

Note the transpose: W_{ij} is the influence of j on i and therefore reads A_{ji}. For a support σ of size m with induced adjacency A_σ, with $J = \mathbf{1}\mathbf{1}^T$ the all-ones matrix,

$$I - W_\sigma \ = \ -\delta I + (1+\delta)J - (\varepsilon+\delta)A_\sigma^T. \tag{4}$$

Every entry is a polynomial in (ε, δ) with integer coefficients. Identity (4), including the transpose, is machine-checked (`closed_form`$^{\llcorner}$).

Definition 2.1. A nonempty $\sigma \subseteq [n]$ *supports a stable fixed point* (SFP) if

(1) $(I - W_\sigma)x = \theta\mathbf{1}$ has a solution with $x > 0$ entrywise;

(2) $\theta + \sum_{j\in\sigma} W_{kj}x_j < 0$ for every $k \notin \sigma$;

(3) every eigenvalue of $-I + W_\sigma$ has negative real part.

If (1)–(3) hold, the zero-extension of x is an equilibrium of (1) near which the vector field is affine with Jacobian spectrum $\operatorname{spec}(-I + W_\sigma) \cup \{-1\}$, so it is linearly, hence locally asymptotically, stable. "Stable" always means linearly stable in this strict sense: points where $I - W_\sigma$ is singular or a Hurwitz determinant vanishes are never counted as stable. When $\sigma = [n]$, condition (2) is vacuous and "target-free clique" reduces to "complete bidirected".

Condition (3) is decided by the Hurwitz determinants of the exact characteristic polynomial: for a real $f(z) = c_0 z^m + \cdots + c_m$ with $c_0 > 0$, $\Delta_k(f) = \det[c_{2i-j}]_{i,j=1}^{k}$ (with $c_l = 0$ outside $0 \le l \le m$), and f has all roots in $\operatorname{Re} z < 0$ iff $\Delta_1, \ldots, \Delta_m > 0$ (Routh–Hurwitz; see Gantmacher, *The Theory of Matrices*, Vol. II, Ch. XV).

2.2 Reparametrisation and legality

Put

$$t = \frac{1-\varepsilon}{\varepsilon}, \qquad \beta = 1 + \frac{\delta}{\varepsilon}, \qquad \gamma = 1 - \frac{1}{\beta}, \qquad q = \frac{\delta(1-\varepsilon)}{\varepsilon} = \frac{(\beta-1)t}{t+1}.$$

The map $(\varepsilon, \delta) \mapsto (\beta, t)$ is a homeomorphism of $(0,1) \times (0,\infty)$ onto $(1,\infty) \times (0,\infty)$ with inverse $\varepsilon = 1/(1+t)$, $\delta = (\beta-1)/(1+t)$, and

$$\text{legal} \iff q > 1 \iff t(\beta - 2) > 1 \tag{5}$$

(`legal_iff`^{L}). In particular every legal point has $\beta > 2$ and $\gamma \in (1/2, 1)$, and along a ray $\delta = (\beta-1)\varepsilon$, $\varepsilon \to 0$, one has $t \to \infty$ and $q \to \beta - 1$. The drive $\theta > 0$ only rescales fixed points: each of (1)–(3), and every boundary event, has the same truth value at every $\theta > 0$ (Lemma G of the reduction note[1]).

Let $N := J - I - A_\sigma^T$ be the *receiver-first non-edge matrix*: $N_{ab} = 1$ iff $a \neq b$ and $\sigma_b \not\to \sigma_a$. Put $C := I + \beta N$ and $K := C + tJ$.

2.3 Rank-one structure

Lemma 2.2 (Rank-one structure; proved)**.** *Let $\varepsilon > 0$. Then*

(i) $I - W_\sigma = \varepsilon K$ *(*`IminusW_eq`^{L}*);*

(ii) $a := \operatorname{adj}(K)\mathbf{1} = \operatorname{adj}(C)\mathbf{1}$ *for every value of t; each $a_i \in \mathbb{Z}[\beta]$ has degree $\le m-1$;*

(iii) $\det K = d + ts$ *with $d = \det C$ and $s = \mathbf{1}^T a$;*

(iv) $\det(\mu I + K) = \phi(\mu) + t\,\omega(\mu)$ *with $\phi(\mu) = \det(\mu I + C)$ and $\omega(\mu) = \mathbf{1}^T \operatorname{adj}(\mu I + C)\mathbf{1}$, where ω has degree exactly $m-1$ in μ and leading coefficient m;*

(v) if $D := \det K \neq 0$, the solution of (1) is $x = (\theta/\varepsilon)\, a/D$, and for $k \notin \sigma$ the drive in (2) equals $\theta\, o_k/D$ with $o_k = d - \beta \sum_b N_{kb} a_b$, where $N_{kb} = 1$ iff $\sigma_b \not\to k$.

Sketch. (i) is an entrywise check using $\varepsilon\beta = \varepsilon + \delta$ and $\varepsilon t = 1 - \varepsilon$. (ii)–(iv) follow from the adjugate form of the matrix determinant lemma, $\det(M + \tau\mathbf{1}\mathbf{1}^T) = \det M + \tau\,\mathbf{1}^T \operatorname{adj}(M)\mathbf{1}$, valid over any commutative ring, and Cramer's rule in column form: in K with column i replaced by $\mathbf{1}$, subtracting t times that column from the others gives C with column i replaced by $\mathbf{1}$. (v) is Cramer's rule together with $W_{kb} = -\varepsilon(t + \beta N_{kb})$. No step divides by $\det C$. Full proofs: `notes/lemmas.md`, §1. □

[1] `notes/lemma_F_reduction.md`, §1.

Lemma 2.3 (Positivity is t-free; proved, ⌞). *If K is positive stable (equivalently, (3) holds) then* $\det K > 0$. *Hence, at every point with $\varepsilon > 0$,*

$$\sigma \text{ is an SFP} \iff a_i(\beta) > 0 \ \forall i, \quad o_k(\beta) < 0 \ \forall k \notin \sigma, \quad \det(\mu I + K) \text{ Hurwitz.}$$

Sketch. $\det K$ is the product of the eigenvalues; real ones are positive and complex ones pair to $|\lambda|^2 > 0$. The Lean proof (`det_pos_of_posStable`, `ctln_pos_iff_adjugate_pos`) argues instead that $x \mapsto \det(xI + K)$ is monic with no root in $[0, \infty)$. □

The Lean development covers $\det K > 0$ and the positivity clause ($x > 0 \iff a > 0$ under stability); the off-support clause $o_k < 0$ is not formalised. Thus in the (β, t) quarter-plane the SFP set is a product of a set of β-intervals (positivity and the off-support condition) with $(0, \infty)$, intersected with the stability set and the legal region; only stability depends on t.

2.4 The $\varepsilon \to 0$ limit

Let $U \in \mathbb{R}^{m \times (m-1)}$ have orthonormal columns spanning $\mathbf{1}^\perp$ and write $X|_\perp := U^T X U$. Let $\alpha := \max \operatorname{Re} \operatorname{spec}(A_\sigma|_\perp)$ be the spectral abscissa of the compressed adjacency; it depends on the graph only. Exactly, $\operatorname{spec} A|_\perp$ can be computed over $\mathbb{Q}$ from any rational basis of $\mathbf{1}^\perp$.

Lemma 2.4 (Limit criterion; proved). *Let $\beta > 1$ and $\gamma = 1 - 1/\beta$.*

(a) $\omega(\mu; \beta) = m \det\big((\mu + 1) I_{m-1} + \beta\, N|_\perp\big)$, *and as $t \to \infty$ one eigenvalue of $K(\beta, t)$ is $mt + O(1)$ while the other $m - 1$ converge to* $\operatorname{spec}\big((1 - \beta) I - \beta A|_\perp^T\big)$.

(b) If $\alpha < -\gamma$ there is T with $K(\beta, t)$ positive stable for all $t > T$; if $\alpha > -\gamma$, then $K(\beta, t)$ is unstable for all large t; $\alpha = -\gamma$ is undecided by the limit. The limit-stable set of legal β is the interval $(2, \beta_c)$, $\beta_c = 1/(1+\alpha)$, nonempty iff $\alpha < -1/2$; always $\alpha \geq -1$, with equality iff $G[\sigma]$ is complete bidirected. The threshold T is uniform on compact subsets of $(2, \beta_c)$.

(c) $a(\beta)$ is one-signed $\iff$ $A^T + \gamma I$ is nonsingular and $w := (A^T + \gamma I)^{-1}\mathbf{1} > 0$; then a is a nonzero multiple of w, and condition (1) holds at (β, t) for some, equivalently every, $t > 0$ iff $w > 0$. For $k \notin \sigma$, $o_k s < 0$ iff $\sum_{b : \sigma_b \to k} w_b < 1$.

(d) If $\alpha < -\gamma$ at a legal β, then $G[\sigma]$ has more than $m(m-1)/2$ edges.

Sketch. (a) Conjugating by the orthogonal matrix $[\mathbf{1}/\sqrt{m} \,|\, U]$ turns tJ into $tm\, e_1 e_1^T$, and the determinant is affine in the $(1,1)$ entry with slope the complementary minor; root continuity (Rouché, or an elementary continuity lemma) gives the eigenvalue statement. (b) follows from (a) and $J|_\perp = 0$, since $N|_\perp = -I - A|_\perp^T$. (c) uses $Ca = d\mathbf{1}$ and $C = -\beta(A^T + \gamma I) + \beta J$, so $(A^T + \gamma I)a = \frac{\beta s - d}{\beta}\mathbf{1}$, together with $\det(A^T + \gamma I) = (-1/\beta)^m (d - \beta s)$. (d) is a trace computation: $\operatorname{tr} A|_\perp = -|E|/m$ and the maximum real part dominates the mean. Full proofs: `notes/lemmas.md`, §4, and `notes/lemma_I_spectral.md`, §§2–6. □

Combining (b) and (c): if $w(\gamma) > 0$ and $\alpha < -\gamma$ at some rational $\beta > 2$, then the full support is an SFP at the legal points $(\varepsilon, \delta) = (1/(1+t), (\beta - 1)/(1+t))$ for all $t > T(\beta)$. We call such a graph *limit-stable*. Lemma 2.4 is the shared dependency of every "for all k" statement in Section 6; it is not formalised in Lean. Parts (b) and (c) are Lemmas I.3 and I.4 of the project notes; they were refereed when the $n = 6$ map was audited and were not re-derived by the referees of the family theorems.

2.5 Reduction to full supports

Lemma 2.5 (Cliques; proved). *Let $0 < \varepsilon < 1$, $\delta > 0$, $\theta > 0$ and let σ be a clique of size m. Then $x = c\mathbf{1}$ with $c = \theta/(\varepsilon + m(1-\varepsilon))$ solves (1), (3) holds, and the drive at $k \notin \sigma$ is $\theta[\varepsilon - r(\varepsilon+\delta)]/(\varepsilon + m(1-\varepsilon))$ where r is the number of $j \in \sigma$ with $j \not\to k$. Hence a clique is an SFP iff it is target-free.*

For a legal point p and $k \geq 2$ let $\mathrm{FS}(k, p)$ be the statement "every k-node digraph whose full support satisfies (1) and (3) at p is complete bidirected".

Theorem 2.6 (Reduction to full supports; proved). *Fix a legal $p = (\varepsilon, \delta)$ and $n \geq 1$. The following are equivalent:*

(i) the target-free clique conjecture holds for every n-node CTLN at p;

(ii) for every n-node G, every SFP support of G at p is a clique;

(iii) $\mathrm{FS}(k, p)$ holds for every $k \in \{2, \ldots, n\}$.

Sketch. (i)⇔(ii) by Lemma 2.5. (iii)⇒(ii): conditions (1) and (3) for σ in G involve only W_σ, which is the weight matrix of the induced subgraph $G[\sigma]$; so a non-clique SFP σ would give a non-complete $|\sigma|$-node graph satisfying (1) and (3) on its full support. (ii)⇒(iii) by padding: given a non-complete H on $k \leq n$ nodes with (1) and (3) on its full support, add $n - k$ nodes receiving no edges from H. At such a node the drive is $\theta - \sum_j c_{kj} x_j$ with all $c_{kj} = 1 + \delta$; subtracting row i of the fixed-point equation shows it is at most $-\delta x_i < 0$, so (2) holds and $V(H)$ is a non-clique SFP of the padded graph. Condition (2) enters only to exclude targets of cliques. Full proof: `notes/lemma_F_reduction.md`, §6 (there called Theorem F). □

Corollary 2.7. *Let $\mathrm{Bad}(n)$ be the set of legal points at which the conjecture fails for some n-node graph, and S_H the set of legal points at which the full support of H is an SFP. Then $\mathrm{Bad}(n) = \bigcup_{k \leq n} \bigcup_H S_H$ over non-complete k-node H, and $\mathrm{Bad}(n) \subseteq \mathrm{Bad}(n+1)$.*

Lemma 2.8 (Openness; proved). *Each S_H is open in the (ε, δ) plane. Consequently every nonempty S_H contains rational points, and at every point of S_H all Hurwitz determinants and coefficients of the characteristic polynomial are strictly positive: a counterexample point is never an "undecided" point.*

We write q_n^* for the supremum of q over legal points at which some non-complete n-node digraph has a stable positive full-support fixed point. By Corollary 2.7, the conjecture for all graphs on at most n nodes holds at every legal point with $q \geq \max_{k \leq n} q_k^*$ and fails at points with q arbitrarily close to that maximum. For $k \leq 4$ there is no counterexample at any legal parameter: $k = 2$ by a direct eigenvalue computation (for a one-way edge $\det K = 1 - t(\beta - 2)$, negative exactly at legal points; a non-adjacent pair is excluded by Lemma 3.8), $k = 3$ by Theorem 4.1 of [4], $k = 4$ is stated as known in [4] (for nondegenerate CTLNs, citing work of Curto, Geneson and Morrison whose exact hypotheses we have not checked), and all three cases are re-certified by our two-parameter engine on all digraphs with at most four nodes; our claims for $k \leq 4$ rest on that certification.[2]

3 The exact machinery

3.1 Boundary sets and three-valued decisions

At a single sampled parameter a vanishing Hurwitz determinant says nothing about the neighbourhood, so every pointwise procedure in this work returns one of three values: *stable*, *not stable*, or *undecided*. An undecided sample is never counted as unstable and never as a counterexample. Regions are decided instead on open cells of a decomposition whose walls are the zero sets of a proved boundary set.

[2] `certificates/region2d_n2to4_all.json`; `notes/lemma_F_reduction.md`, §§5, 8.

Lemma 3.1 (Imaginary-axis roots; proved). *If f is real of degree $m \geq 2$ and $f(iw) = 0$ for some real $w \neq 0$, then $\Delta_{m-1}(f) = 0$.*

Sketch. With $P(x) = x^m f(1/x)$, the factor $1+w^2x^2$ divides $P = (1+w^2x^2)R$; for $V(x) = xR(-x) \neq 0$ the product PV is odd, and its even coefficients are exactly the entries of $H_{m-1}v$ where H_{m-1} is the matrix whose determinant is Δ_{m-1}. This is the special case of Orlando's formula that is needed; it is proved self-contained in `notes/lemmas.md`, §2. □

Lemma 3.2 (Boundary set at fixed t; proved). *Fix $t > 0$ and let $p_t(\mu; \beta) = \det(\mu I + K)$. Let P_t be the family $a_1, \ldots, a_m$, $D_t = d + ts$, o_k ($k \notin \sigma$), $\Delta_{m-1}(p_t)$, all in $\mathbb{Q}[\beta]$. If some member is identically zero, σ is an SFP at no legal β. Otherwise the SFP verdict is constant on each component of the legal β-range minus the real roots of P_t, and false at those roots. The same holds with β fixed and t varying, and in the limit $t = \infty$ with ω, s, $\Delta_{m-2}(\omega)$ in place of p_t, D_t, $\Delta_{m-1}(p_t)$.*

Sketch. On a component, every member of P_t has constant sign; by Lemma 3.1 no root of p_t meets the imaginary axis; by continuity of roots the number of roots in the right half-plane is locally constant. `notes/lemmas.md`, §2. □

The two-parameter version (Lemma B′ and Corollaries B″, B‴ of the project notes[3]) shows that the verdict is invariant on connected components of the complement of the curves $\{a_i = 0\}$, $\{D = 0\}$, $\{\Delta_{m-1} = 0\}$ and the legality boundary $L(\beta, t) = t(\beta - 2) - 1 = 0$ in the open legal region, and that over each open β-interval free of the roots of a projection set (leading coefficients, discriminants and pairwise resultants of these polynomials in t) the sectors of a cylindrical algebraic decomposition are sign-invariant cells. Each full-dimensional cell is decided at one exact rational sample; lower-dimensional cells are boundary points and are never SFP.

3.2 The engines

One-parameter engine. At fixed t (or $t = \infty$), the polynomials of Lemma 3.2 are obtained by exact integer evaluation and interpolation with provably sufficient degree bounds, their real roots are isolated by continued fractions into irreducible minimal polynomials with rational isolating intervals, and each cell is decided at an exact sample by the Hurwitz routine. Every isolation is replayed by Sturm counts and every sample is cross-checked against an independent pointwise verifier. The engine agrees with an independent symbolic path on all 7,194 (graph, support, ε) cases at $n = 4$ and on 900 at $n = 5$.[4]

Two-parameter engine. A cylindrical algebraic decomposition of the legal (β, t) plane on the proved boundary set, restricted to the positivity cells, with a Descartes-bisection replay of every root count that is independent of the isolator. It reproduces the theorem for $k \leq 4$ on all digraphs with at most four nodes and agrees with an earlier implementation on all 706 five-node survivors of a weaker filter.[5]

Policy. A cell whose Hurwitz test is undecided at three samples is *escalated*, not decided; there were 0 escalations at $n = 5, 6, 7$. By Lemma 2.8 undecided points cannot hide counterexamples.

3.3 Ray reduction to the legality curve

Lemma 3.3 (Ray identity; proved, ^{L}). *$K(\lambda\beta, \lambda t) = (1 - \lambda)I + \lambda K(\beta, t)$. Hence if $K(\beta, t)$ is positive stable, so is $K(\lambda\beta, \lambda t)$ for every $\lambda \in (0, 1]$ (`affine_identity`, `posStable_ray`^{L}).*

[3] `notes/lemma_Bprime_2d.md`.

[4] `certificates/validate_fastregion_n4.json`, `certificates/validate_fastregion_n5.json`.

[5] `certificates/region2d_n2to4_all.json`, `certificates/region2d_v2_validation.json`.

Lemma 3.4 (Ray reduction; proved, forward direction ^{L}). *The full support is stable at some legal (ε, δ) if and only if $K(2+1/t,\, t)$ is positive stable for some $t > 0$. Moreover, if the full support is an SFP at a legal (β_0, t_0), the corresponding curve point has $\beta' < \beta_0$.*

Proof. ($\Leftarrow$) Positive stability is open, so $K(2 + 1/t + \eta, t)$ is stable for small $\eta > 0$, and that point is legal. ($\Rightarrow$) Let (β, t) be legal and stable and $f(\lambda) = \lambda t(\lambda\beta - 2) - 1$. Then $f(2/\beta) = -1 < 0 < f(1)$, so f has a root $\lambda^* \in (2/\beta, 1)$, explicitly $\lambda^* = (t + \sqrt{t^2 + \beta t})/(\beta t)$; the point $(\lambda^*\beta, \lambda^* t)$ lies on the curve $t(\beta - 2) = 1$ and is stable by Lemma 3.3. The forward direction is `lemmaR_forward_ctln`^{L}; the converse needs continuity of eigenvalues and is not formalised. □

Since positivity is t-free (Lemma 2.3), a graph that is never stable on the curve has no SFP anywhere in the legal plane. Along the curve, $tK(2+1/t, t) = L(t) := tI + (2t+1)N + t^2 J$ is an integer polynomial matrix; the coefficient of μ^{m-k} in $\det(\mu I + L(t))$ has t-degree at most $k + 1$, and by Lemma 3.2 with t as the parameter the verdict changes only at positive roots of $\det L(t)$ and of $\Delta_{m-1}(t)$. One exact sample per root-free interval decides the curve (the *curve test*). A fast path discards a graph when no $t > 0$ makes every coefficient positive (Stodola's condition, below). The curve test takes about 35–45 ms per seven-node graph, and on all 9,608 five-node digraphs returns 95 stable and 9,513 never, each of the 95 with an exact legal witness.[6]

3.4 Necessary conditions valid at every parameter

Every prune below is a theorem valid at *every* legal parameter, and all five are machine-checked. The twin obstruction is not a separate stage of the $n = 7$ chain.

Lemma 3.5 (Domination; proved, ^{L}). *Node k* graphically dominates *$j \neq k$ if $j \to k$, $k \not\to j$, and every $i \notin \{j, k\}$ with $i \to j$ also has $i \to k$ [2, Def. 1.1]. If k dominates j, then every fixed point of (1) with uniform drive has $x_j = 0$, for every $\varepsilon > 0$ and $\delta \geq 0$; in particular the full support has no positive fixed point. The same holds for domination inside an induced subgraph $G[\sigma]$ (`lemmaD`, `lemmaD_full_support`, `lemmaD_induced`^{L}).*

Sketch. For $x \geq 0$ one has $y_j(x) - y_k(x) \leq (1 - \varepsilon)x_j - (1 + \delta)x_k$, where y_l is the input to node l. At a fixed point with $x_j > 0$ this forces $y_k > 0$ and then $0 \geq \varepsilon x_j + \delta x_k > 0$. Neither $\varepsilon < 1$ nor legality is used, but the uniform drive is: with $\theta_j > \theta_k$ the conclusion fails. `notes/lemmas.md`, §5. □

Lemma 3.6 (Edge bound; proved, ^{L}). *Let B, N_n, O be the numbers of bidirected, non-adjacent and one-way unordered pairs of an m-node graph. The sum of the 2×2 principal minors of K is*

$$e_2(K) = B(1 + 2t) + N_n(1 - \beta)(1 + \beta + 2t) + O\big(1 - t(\beta - 2)\big).$$

If K is positive stable at a legal point then $e_2(K) > 0$, hence $B > N_n$ and the graph has at least $m(m-1)/2 + 1$ edges (`e2_closed_form`, `edge_bound_ctln`^{L}).

At a legal point the non-adjacent and one-way terms are negative, which gives $B > N_n$. In Lean the spectral step (positive stability forces $e_2 > 0$) is proved from Mathlib's description of characteristic-polynomial coefficients as sums of principal minors and the Gauss–Lucas theorem. For $n = 7$ the bound reads "at least 22 edges".

Lemma 3.7 (Stodola; proved, ^{L}). *A monic real polynomial with all roots in $\operatorname{Re} z < 0$ has all coefficients positive. Hence if K is positive stable, every coefficient of $\det(\mu I + K)$ is positive (`stodola`, `coeff_pos_of_posStable`, `ctln_coeff_pos`^{L}).*

[6] `n7/validate_curve_n5.json`; `notes/lemma_M_refuted.md`, §3.

Lemma 3.8 (Twin obstruction; proved, ⌞). *If $u, v \in \sigma$ are non-adjacent and have the same in-neighbours in $\sigma \setminus \{u, v\}$, then $e_u - e_v$ is a left eigenvector of W_σ with eigenvalue $1 + \delta$, so $-I + W_\sigma$ has the eigenvalue $\delta > 0$ and σ is never stable, for every ε and every $\delta > 0$ (`twin_not_hurwitz`, `twin_not_hurwitz_induced`⌞). For bidirected in-twins the eigenvalue is $-\varepsilon$.*

Together with Lemma 2.3 (a positive stable full support needs a legal β with all $a_i(\beta) > 0$) and the density bound Lemma 2.4(d) for the limit, these are the only prunes used. An early placeholder for domination was replaced by the verbatim definition of [2] before any sweep used it.

4 Complete maps for $n \leq 6$

4.1 Validation gates

Two published results were reproduced before any new computation was trusted.

Geneson's counterexample. The paper [4] gives its graph as a receiver-first non-edge matrix; our adjacency matrix is its complement and transpose, and the weight convention is the same. The printed integer certificate $Cv = 179298949 \cdot \mathbf{1}$ reproduces exactly, our fixed point is proportional to the paper's v, and the full support is a stable non-clique fixed point at $\varepsilon = 1/100$, $\delta = 29/2500$ and at $\varepsilon = 1/1000$, $\delta = 29\varepsilon/25$, while it is a positive but unstable fixed point at $\varepsilon = 1/10, 1/20, 1/50$ on the same ray.[7]

The published $n = 5$ classification (the package accompanying Curto's $n = 5$ data, which stores the transpose of our adjacency matrix). At its default parameters $\varepsilon = 1/4$, $\delta = 1/2$, every recorded fixed-point support of every one of the 9,608 five-node digraphs matches our exact verifier: 24,442 supports, 0 disagreements, 0 boundary cases (the package records no stability data; our verifier finds 14,488 of the supports stable). Under the wrong (untransposed) convention a sampled version of this check fails on 27 of 30 graphs.[8]

Our enumerations reproduce the number of digraphs up to isomorphism (OEIS A000273): 9,608 for $n = 5$, 1,540,944 for $n = 6$ and 882,033,440 for $n = 7$.

4.2 $n \leq 5$

Theorem 4.1 ($n \leq 5$; certified)**.** *No non-complete digraph on at most five nodes has a stable positive full-support fixed point at any legal parameter. Consequently, by Theorem 2.6, the target-free clique conjecture holds for every CTLN on at most five nodes at every legal parameter.*

Of the 9,608 five-node digraphs, 390 admit a legal β with all $a_i(\beta) > 0$; the two-parameter decomposition certifies 389 of them as never stable and only K_5 as stable, with 0 escalations. (An earlier, weaker common-sign filter left a superset of 706, which was also certified in full: 705 never, K_5 stable.)[9] This agrees with the independent result of [1].

4.3 $n = 6$

Theorem 4.2 ($n = 6$; certified, hand-verified)**.** *Among the* 1,540,944 *digraphs on six nodes, exactly three non-complete graphs have a stable positive full-support fixed point at some legal parameter (Table 1). Every other six-node graph is certified never stable and positive at every legal (ε, δ).*

[7] `certificates/stage0_geneson_2607.21396.json`; a permanent test in the project's test suite.
[8] `certificates/stage1_n5_diff.json`.
[9] `certificates/region2d_n5_shard0of1_tally.json`, `certificates/n5_summary.json`.

graph	adjacency rows	edges	limit-stable q	relation
G_1 (index 1,000,255)	`011000 000011 000111 110011 110101 101110`	19	$(1, q_6^*)$	Geneson
G_2 (index 1,424,629)	`011000 000101 000111 100011 110101 111110`	19	$(1, q_6^*)$	switch of G_1
H_3 (index 1,387,528)	`011000 000011 000111 110010 100101 111110`	18	$(1, q_c')$	new

Table 1: The three six-node counterexamples, in canonical labelling (row i lists the j with $i \to j$); G_2 is a bidirected in-twin switch of G_1. $q_c' \approx 1.0830$ is the root in $(1, 13/12)$ of $q^5 + 2q^4 + q^3 - 2q^2 - 2q - 1$.

The chain. In the limit sweep over all 1,540,944 graphs, 1,512,594 have no legal β with all $a_i > 0$ and therefore (Lemma 2.3) no stable positive full support at any parameter; 28,350 strict survivors remain, all among the 204,378 domination-free graphs; exactly four are limit-stable (K_6, G_1, G_2, H_3); 0 escalations. The two-parameter decomposition was run on a superset of 34,071 survivors (an earlier weaker filter): 34,067 never stable anywhere, 4 stable somewhere, 0 escalations, 0 errors, every cell cross-checked pointwise and every root count replayed. Exact β-slices at $\varepsilon = 1/1000$ and $1/100$ show K_6 and the three graphs; at $\varepsilon = 1/10$ only K_6 and Geneson's class; at $\varepsilon = 1/4$ and $1/2$ only K_6.[10]

The regions. For G_1 and G_2 the stable positive region is $\{2 < \beta < \beta_c,\ t > \max(1/(\beta-2), T(\beta))\}$ with $T(\beta) < 1/(\beta-2)$ on an initial interval $(2, \beta_1)$, β_1 a degree-14 algebraic number in $(17/8, 15/7)$: these graphs are stable at *every* legal t when δ/ε is close enough to 1, so counterexamples exist with q arbitrarily close to 1. For H_3 the two thresholds coincide: the full support is an SFP iff $2 < \beta < \beta_1'$, at every legal t, where $\beta_1' \approx 2.08302$ is the root in $(152/73, 25/12)$ of $\beta^5 - 3\beta^4 + 3\beta^3 - 3\beta^2 + 2\beta - 1$, a positivity boundary ($a_5 = 0$). A witness is $\varepsilon = 1/101$, $\delta = 27/2626$, $q = 1350/1313$.[11]

Twins. G_2 is obtained from G_1 by moving one out-edge between a pair of bidirected in-twins.

Proposition 4.3 (Bidirected in-twin switching; proved)**.** *Let $u \leftrightarrow v$ be bidirected and have the same in-neighbours outside $\{u, v\}$, and let G' be obtained from G by replacing an edge $u \to w$ (with $v \not\to w$) by $v \to w$. Then for every support, every $\varepsilon, \delta > 0$ and every θ, G and G' have the same fixed points (same vectors) and the same Jacobian characteristic polynomials.*

More generally, two graphs have identical full-support data for all parameters iff after relabelling they are cospectral and have the same in-walk counts $(A^T)^k \mathbf{1}$ for all k; the 34,071 six-node survivors form 21,800 such classes. Bidirected in-twin switching preserves all supports, whereas other members of a class can differ in their proper supports.[12] So modulo twin switching there are exactly two six-node counterexample graphs.

4.4 The threshold at $n \le 6$

In the limit, $\beta_c(G_1) = -1/\operatorname{Re}\nu_1$, where ν_1 is a root of the quartic factor $6z^4 + 11z^3 + 16z^2 + 10z + 3$ of the characteristic polynomial of the compressed non-edge matrix (the other factor is z, from a pair of twins). Real parts of roots of a quartic are roots of a polynomial of degree at most $\binom{4}{2} = 6$; here β_c is a root of the irreducible sextic $797b^6 - 6468b^5 + 22544b^4 - 44440b^3 + 53280b^2 - 38016b + 13824$, and the substitution $b = q + 1$ gives the sextic in q below.

Theorem 4.4 (Threshold for $n \le 6$; certified)**.** *Let q_6^* be the root in* $(561/481,\ 7/6)$ *of*

$$797q^6 - 1686q^5 + 2159q^4 - 3004q^3 + 2499q^2 - 2158q + 1521 \qquad (q_6^* \approx 1.1664).$$

[10] `certificates/n6_summary.json`; `certificates/region2d_n6_shard{0,1,2}of3*`; `certificates/region_n6_eps*`.

[11] `certificates/COUNTEREXAMPLE_CANDIDATE_n6_idx1387528.json`, `certificates/COUNTEREXAMPLE_CANDIDATE_n6_idx1424629.json`.

[12] `notes/lemma_J_equivalence.md`; `certificates/n6_survivor_equivalence_classes.json`.

For every graph on at most six nodes, the target-free clique conjecture holds at every legal parameter with $q \geq q_6^*$; for every $q \in (1, q_6^*)$ there is a legal parameter with that value of q at which it fails for a six-node graph. The supremum is approached by G_1 and G_2 as $t \to \infty$ and is not attained.

Sketch. By Theorem 4.1, Theorem 4.2 and Corollary 2.7, $\mathrm{Bad}(6) = S_{G_1} \cup S_{G_2} \cup S_{H_3}$. On every finite-$t$ cell $q = (\beta - 1)t/(t+1) < \beta - 1$, and the exact suprema over the certified cells are computed by the threshold tool from the two-parameter certificates; the limit interval of G_1 starts at $q = 1$, so the union covers $(1, q_6^*)$.[13] □

Geneson's bound (2) at $n = 6$ reads $q \geq 4 - 3\varepsilon/2$; Theorem 4.4 shows the true value is $q_6^* \approx 1.1664$. The same sextic root, $1.16635502785\ldots$, was obtained independently in [1].

5 The complete map at $n = 7$

5.1 The accounting chain

Theorem 5.1 ($n = 7$; certified, hand-verified)**.** *Among the* 882,033,440 *digraphs on seven nodes, exactly* 124 *non-complete graphs have a stable positive full-support fixed point at some legal (ε, δ). Each carries an exact two-parameter region certificate. Of the* 124*, exactly* 123 *are limit-stable; the remaining graph,* `&FNFL~[udo?`*, is stable only at finite ε.*

Every digraph is either removed by a necessary condition valid at every legal parameter, or certified by the two-parameter engine (Table 2). Each stage was validated before use against the $n = 6$ map (the positivity stages reproduce the 28,350 six-node strict survivors exactly, with zero difference in both directions; the curve test applied to those survivors returns exactly the four stable six-node graphs).

stage	condition (where proved)	removed	remaining
0	enumeration (OEIS A000273)		882,033,440
1	a dominated node (Lemma 3.5)	764,800,518	117,232,922
2	no $\beta > 2$ with all $a_i(\beta) > 0$, exact Descartes bounds (Lemma 2.3)	107,600,789	9,632,133
3	undecided lines of stage 2 resolved by exact root isolation	723,010	8,909,123
4	at most 21 edges (Lemma 3.6)	5,135,775	3,773,348
5	never stable on the legality curve (Lemma 3.4)	3,773,183	165
6	stable somewhere, never stable and positive together (CAD)	40	125

Table 2: Accounting chain at $n = 7$. The 125 survivors are K_7 and the 124 counterexamples; 0 escalations at every stage.

Of the 9,632,133 lines kept at stage 2, 8,845,537 were certified positive and 786,596 were undecided by the fast exact filter; stage 3 resolved the latter exactly. The single run performing stages 4 and 5 over the 8,909,123 positivity survivors took 16.4 CPU-hours (6.6 ms per line).[14] Independently, the limit classification was run first: a spectral prefilter over all 882,033,440 digraphs left 638,042 survivors (1,889 strict), whose exact limit classification gave the 123 limit-stable non-clique graphs and K_7.[15] The finite-ε map contains all of them, as Lemma 2.4 requires.

[13] `certificates/qstar/qstar_n6.json`, tool `n7/qstar.py`.

[14] `certificates/n7_finite_summary.json` (field `chain`, `accounting_ok = true`); `certificates/n7_curve_TOTAL.json`; `certificates/n7_region2d/`; runbook `n7/REPRODUCE_n7_finite.md`.

[15] `certificates/COUNTEREXAMPLE_CANDIDATES_n7_limit.jsonl`, `certificates/n7_limit_tally.json`. For 15 graphs the first stored pointwise witness in the limit file is an illegal point ($q \leq 1$); use `certificates/n7_candidates_legal_witnesses.json`.

5.2 The 124 graphs

- Edge counts range from 23 to 31 of 42 possible (counts $7, 7, 18, 10, 6, 22, 34, 17, 3$ for $23, \ldots, 31$ edges); all satisfy the edge bound (≥ 22) and all are domination-free.
- They form 91 classes of identical full-support data (Section 4): 71 singletons, 16 pairs, two classes of 4, one of 5 and one of 8; 64 graphs carry a bidirected in-twin pair.
- Of the 123 limit-stable graphs, the limit critical values q_c (right ends of the limit-stable q-intervals) have 69 distinct minimal polynomials; the most frequent is $q^5+2q^4+q^3-2q^2-2q-1$ (17 graphs, the same quintic as for H_3), and the largest q_c is the plastic number, the real root of q^3-q-1 (6 graphs). For 6 of the 123 the positivity set starts strictly above $\beta = 2$.
- 21 of the 123 contain a six-node counterexample as an induced subgraph; none contains K_6.

Every graph comes with a legal exact witness point.[16]

Corollary 5.2 (certified)**.** *Every seven-node graph whose full support is an SFP at some legal parameter has $\alpha \leq -1/2$, with equality only for* `&FNFL~[udo?`.

5.3 A counterexample invisible in the limit

The graph E_0 = `&FNFL~[udo?` (28 edges) has adjacency rows `0011110 0011100 1101111 1110111 0011011 0100101 1100000`. Its compressed adjacency satisfies

$$7\chi_{A|_\perp}(\lambda) = (\lambda+1)(\lambda^2+\lambda+1)(7\lambda^3+14\lambda^2+14\lambda+5),$$

so $\alpha = -1/2$ exactly and, by Lemma 2.4(b), the graph is limit-stable at no legal β. Nevertheless its full support is a stable positive fixed point at $\varepsilon = 1/49$, $\delta = 33/1568$ ($q = 99/98$) and at $\varepsilon = 1/36$, $\delta = 17/588$. Its stable region is bounded in t at every β: on its single positivity cell $2 < \beta < \beta_1'$ ($\beta_1' \approx 2.08302$, the quintic root of Table 1, isolated here in $(27/13, 25/12)$), at the sample $\beta = 53/26$, the stable range is $26 < t < t_1$ with t_1 an algebraic number in $(53, 54)$, and the full support is unstable for larger t. The supremum of q over its region is an algebraic number of degree 25 in (685003/657781, 2170535/2084278), about 1.0414, approached at the corner where $\Delta_6 = 0$ meets the positivity boundary, not as $t \to \infty$.[17] Structurally, E_0 is H_3 with one node doubled into a closed-twin pair, and the factor $\lambda^2+\lambda+1$ is a *moving* eigenvalue pair (not an eigenvalue of A) that sits exactly on the boundary line; see Theorem 6.5.

5.4 The limit does not determine the finite map

Write $c = 1 + t/\beta > 1$. A one-line computation gives the identity

$$K = \beta\,(cJ - A^T - \gamma I), \tag{6}$$

so K is positive stable iff $\nu_A(c) := \min \mathrm{Re}\,\mathrm{spec}(cJ - A) > \gamma$, and legality is $\gamma > c - \sqrt{c^2 - c}$, which is the stability boundary of a single one-way edge. As $c \to \infty$, $\nu_A(c) \to -\alpha$. The natural hope is that ν_A is nondecreasing, i.e. that stability at a legal finite t implies stability for all larger t at the same β ("Lemma M").

[16] `certificates/COUNTEREXAMPLE_CANDIDATES_n7_finite.jsonl`; `certificates/n7_candidates_legal_witnesses.json`.

[17] `certificates/COUNTEREXAMPLE_CANDIDATE_n7_finite_FNFL.json`; `certificates/qstar/qstar_fnfl.json` (the exact bracket is recorded there and in `qstar_n7_finite.json`; the file's aggregate block is stale and reports only a weaker sample bound).

Proposition 5.3 (Lemma M is false; certified). *(a) For A with rows `011 001 100` and $\beta = 22/5$, K is positive stable at the legal point $t = 11/25$ and not at $t = 1$ or $t = 100$; here $\alpha = -2/3 > -17/22 = -\gamma$. (b) For the domination-free seven-node graph `&FK@dyUfXp?` (one of the 123) at $\beta = 10561/5000$, the full support is an SFP at $t = 10$ and positive but unstable at $t = 1000$, and the limit polynomial is not Hurwitz.*

Example (a) is not a CTLN counterexample (it fails positivity), but (b) shows the failure persists with positivity. In the latter, the right end of the stable β-interval at $t = 10, 30, 100, 1000, 10^4$ lies strictly above the limit value β_c.[18] Lemma 2.3 shows that every such failure under positivity is an imaginary-pair crossing: on the positivity set $s > 0$, so $D = d + ts$ increases in t and a real eigenvalue crosses 0 only upward.

5.5 The bulge criterion

Let $\chi(z) = \det(zI + A)$, $\psi(z) = \mathbf{1}^T \operatorname{adj}(zI + A)\mathbf{1}$ and $\hat{s} = \psi/\chi = \mathbf{1}^T(zI + A)^{-1}\mathbf{1}$. Then $\det(zI - (cJ - A)) = \chi(z) - c\psi(z)$; roots of $\gcd(\chi, \psi)$ are eigenvalues for every c (*fixed*), the others are *moving*.

Theorem 5.4 (First-order asymptotics; proved). *Assume $\nu_\infty := -\alpha \in (1/2, 1)$ and that every root z_0 of ψ with $\operatorname{Re} z_0 = \nu_\infty$ is simple and not a root of χ. Let $R^* = \min \operatorname{Re}\big(1/\hat{s}\,'(z_0)\big)$ over these roots. Then $\nu_A(c) = \nu_\infty + R^*/c + O(c^{-2})$, and the boundary of the stable region near (β_c, ∞) is $\beta = \beta_c + \beta_c^3 R^*/t + O(t^{-2})$.*

(a) If $R^* > 0$, then for every β slightly above β_c the stable t-set is nonempty and bounded above; if moreover $a(\beta_c) > 0$, these points are SFPs.

(b) If $R^* < 0$, the stable t-set is upward closed for large t near β_c.

Sketch. With $e = 1/c$ the moving eigenvalues solve $\psi_1(z) - e\chi_1(z) = 0$ ($\chi = h\chi_1$, $\psi = h\psi_1$, $h = \gcd$); the implicit function theorem gives an analytic branch $z(e) = z_0 + e\,\chi_1(z_0)/\psi_1'(z_0) + O(e^2)$, and $\chi_1(z_0)/\psi_1'(z_0) = 1/\hat{s}\,'(z_0)$. `notes/lemma_M_refuted.md`, §4. □

Proposition 5.5 (Bulge rule at $n = 7$; certified). *For all 124 graphs, R^* was computed exactly (51 positive, 73 negative, none zero; every critical root simple and not shared with χ). A graph has a stable cell bounded above in t if and only if $R^* > 0$ and its fixed point is positive at β_c; this holds for exactly 40 graphs, and every bounded cell starts exactly at β_c. The 11 graphs with $R^* > 0$ that lose positivity before β_c do not bulge as fixed points. For E_0, $R^* = 1/4$ and $\beta_c = 2$.*[19]

At $n = 6$ no stable cell is bounded in t (Geneson's class has $R^* < 0$, and the limit interval of H_3 ends at a positivity root), so the $\varepsilon \to 0$ classification describes the six-node regions faithfully, while it understates the regions of 40 of the seven-node graphs and misses E_0 entirely.

5.6 The threshold at $n \le 7$

Theorem 5.6 (Threshold for $n \le 7$; certified, hand-verified). *Let $\rho \approx 1.3247$ be the plastic number, the real root of $q^3 - q - 1$ (isolated in $(45/34, 4/3)$). Then $q_7^* = \rho$, and the supremum is not attained. Consequently, for every graph on at most seven nodes the target-free clique conjecture holds at every legal parameter with $q \ge \rho$, and for every $q \in (1, \rho)$ there is a legal parameter with that value of q at which it fails for a seven-node graph.*

[18] `notes/lemma_M_refuted.md`, §2; scripts in `notes/prover_scripts_M/`.

[19] `certificates/n7_bulge_Rstar_exact.json`; `notes/bulge_exact/`.

Sketch. On every finite cell $q < \beta - 1$. On the 258 stable cells unbounded in t the supremum is $\beta_{\mathrm{hi}} - 1$ with β_{hi} a limit endpoint, and the largest is $1 + \rho$ (six graphs, whose limit interval ends at the positivity root $\beta^3 - 3\beta^2 + 2\beta - 1 = 0$, i.e. $q^3 - q - 1 = 0$, with the stability boundary further right). On the 40 bounded cells the outer isolating bound gives $\beta_{\mathrm{hi}} \le 16/7$, so $q < 9/7 < \rho$. Four of the six plastic-number graphs have limit intervals starting at $q = 1$, so the union of regions covers $(1, \rho)$. Combine with Theorem 4.4 and Corollary 2.7. An independent run of the exact threshold tool over all 298 stable cells returns matching upper and lower bounds with no undecided cell.[20] □

Geneson's bound gives $q \ge 5 - 2\varepsilon$ at $n = 7$; the true threshold is the plastic number. The clique-class family of Theorem 6.3 alone reaches only about 1.2747 at $n = 7$, and other graphs approach the supremum.

6 Infinite families

The theorems in this section are proved for every member of a family; tables and individual instances are certified only. Instances on $n \ge 8$ nodes are certified by the pointwise exact verifier but have *not* been hand-verified, and are reported as candidates in that sense. The proofs use the base graphs only as explicit matrices. Theorems 6.3, 6.5, 6.6 and 6.7 rest on Lemma 2.4(b),(c) (Lemmas I.3 and I.4 of the notes: transfer from the limit to finite t, and t-free positivity); Theorem 6.8 uses only part (c).

6.1 The non-edge picture and blow-ups

Let $D := J - I - A$ be the sender-first non-edge matrix ($D = N^T$ for the full support) and $\psi_D(z) := \mathbf{1}^T \operatorname{adj}(zI - D)\mathbf{1} = m\,\chi_{D|_\perp}(z)$. Since $A|_\perp = -I - D|_\perp$, Lemma 2.4 reads: *the full support is limit-stable at β iff every root of ψ_D has real part $> -1/\beta$*; and the fixed point is positive iff $u := (D^T + \beta^{-1} I)^{-1}\mathbf{1} > 0$ and $\mathbf{1}^T u > 1$, in which case $w = u/(\mathbf{1}^T u - 1)$.

Lemma 6.1 (Closed-twin blow-up; proved)**.** *Let H have m nodes and let $\mu \in \mathbb{Z}^m_{>0}$, $M = \operatorname{diag}(\mu)$. Let H^μ replace node a by a bidirected clique of μ_a nodes, with every node of class a sending to every node of class $b \ne a$ iff $a \to b$ in H; $n = \sum \mu_a$. Then*

$$n\,\chi_{D'|_\perp}(z) = z^{n-m}\,\mu^T \operatorname{adj}(zI - DM)\mathbf{1},$$

and for every γ with $A'^T + \gamma I$ nonsingular the positivity vector of H^μ is constant on classes, $w' = Ey$ with $\big((A^T + I)M + (\gamma - 1)I\big)y = \mathbf{1}$, where E is the class-indicator matrix.

Sketch. $D' = EDE^T$ and $E^T E = M$; push-through $(zI - EDE^T)^{-1}E = E(zI - DM)^{-1}$ and Sylvester's identity $\det(zI_n - EDE^T) = z^{n-m}\det(zI_m - DM)$.[21] □

For a base G, in-set S and out-set T, let $G[S, T; k]$ add a bidirected clique V of k nodes, with $i \to v$ for all $v \in V$ iff $S_i = 1$ and $v \to i$ iff $T_i = 1$; put $b = \mathbf{1} - S$, $c = \mathbf{1} - T$.

Lemma 6.2 (Clique-class extension; proved)**.** *(X1) $n\,\chi_{D'|_\perp} = z^{k-1}\Phi_k$ with $\Phi_k(z) = z\,\psi_D(z) + k\,\chi_L(z)$, $L := D - b\mathbf{1}^T - \mathbf{1}c^T$. (X2) If T is all of G, then for every $k \ge 1$ and rational $\beta > 1$ the full support of $G[S, T; k]$ is positive iff $w_G(\beta) > 0$ and $\sigma_S(\beta) := \sum_{i \in S} w_{G,i}(\beta) > 1$; this does not depend on k or t. (X3) As $k \to \infty$ the moving spectrum converges to $\operatorname{spec}(L)$. With several sink classes of sizes k_j and in-sets S_j, the same holds with L replaced by $D - \bar{b}\mathbf{1}^T$, $\bar{b}$ the size-weighted mean of the b_j, and positivity holds iff $w_G > 0$ and $\sigma_{S_j} > 1$ for every j.*

[20] `certificates/qstar/qstar_n7_finite.json`; `certificates/qstar/qstar_n7limit.json`.

[21] `notes/research_wf3/theory/notes_theory.md`, §2.

6.2 Counterexamples on every $n \geq 6$

Let G_1 be Geneson's graph in the paper's labelling (rows `011011 101011 110101 111000 011000 000110`), $S = (1,1,1,1,1,0)$, and $F_k := G_1[S, \mathrm{all}; k]$, $n = 6 + k$.

Theorem 6.3 (Clique-class family; proved, refereed). *For every $k \geq 0$, $\alpha(F_k|_\perp) < -1/2$, and F_k is positive at every $\beta \in (2, b_p)$, where $b_p - 1 = \rho$ is the plastic number. Hence for each k and each rational $\beta \in (2, \min(b_p, \beta_c(k)))$ the full support of F_k is a stable positive non-clique fixed point at $(\varepsilon, \delta) = (1/(1+t), (\beta-1)/(1+t))$ for all large t. In particular the target-free clique conjecture fails for every $n \geq 6$.*

Sketch. By Lemma 6.2, $\Phi_k = z^2 R_k$ with $R_k = (6+k)z^4 + (11+k)z^3 + (16+k)z^2 + 10z + 3$. The shifted polynomial $16R_k(-\frac{1}{2} - x) = 16(k+6)x^4 + 16(k+1)x^3 + 8(2k+17)x^2 + 12(k+1)x + (3k+16)$ has Hurwitz determinants $16(k+1)$, $64(k+1)(k+16)$, $8192(k+1)^2$ and $8192(k+1)^2(3k+16)$, all positive for $k > -1$. Positivity: the components of $\mathrm{adj}(\beta A^T + (\beta-1)I)\mathbf{1}$ and $\sigma_S - 1$ are explicit polynomials; exact root counting on $[2, 93/40]$ shows a single sign change, at the root b_p of $b^3 - 3b^2 + 2b - 1$.[22] □

The limit supremum of q along the family is $\min(b_p, \beta_c(k)) - 1$: about 1.1664 ($k = 0$), 1.2747 ($k = 1$), the plastic number 1.3247 for $k = 2, \ldots, 8$ ($n = 8, \ldots, 14$), then decreasing to 1 (since $\chi_L = z^4(z^2 + z + 1)$ has roots on $\mathrm{Re}\, z = -1/2$). The same construction proves 9 families on the three six-node graphs and 617 families on 117 of the 124 seven-node graphs, each for every $k \geq 1$, by exact real-root isolation in k; an independent referee re-proved all 626 for every real $k \geq 1$.[23]

6.3 Finite-ε-only counterexamples on every $n \geq 7$

Lemma 6.4 (Boundary mechanism; proved). *Suppose (i) ψ has a conjugate pair of simple roots $z_0, \bar{z}_0$ with $\mathrm{Re}\, z_0 = 1/2$ and $\chi(z_0) \neq 0$; (ii) every other root of ψ has real part $> 1/2$; (iii) $R := \mathrm{Re}\big(\chi(z_0)/\psi'(z_0)\big) > 1/8$; (iv) $w(2) = (A^T + \frac{1}{2}I)^{-1}\mathbf{1} > 0$. Then $\alpha = -1/2$, so the graph is never limit-stable at a legal β, yet its full support is a stable positive fixed point at legal points with ε arbitrarily small (and $\beta \to 2^+$).*

Sketch. By Theorem 5.4, $\nu_A(c) = \frac{1}{2} + R/c + O(c^{-2})$, while the legality boundary is $c - \sqrt{c^2 - c} = \frac{1}{2} + \frac{1}{8c} + O(c^{-2})$. If $R > 1/8$ there is room between them for all large c; choose a rational γ in between. Positivity holds near $\beta = 2$ by (iv) and is t-free. □

Let $H =$ `&FNDL~[uds?` (one of the 123) and let E_a be its blow-up with class sizes $(1, 1, 2, a, 1, 1, 1)$, $n = 7 + a$; E_0 is `&FNFL~[udo?`.

Theorem 6.5 (Finite-ε-only family; proved, refereed). *For every $a \geq 0$, E_a is not a clique, $\alpha(E_a|_\perp) = -1/2$ exactly, and the full support of E_a is a stable positive fixed point at legal points with ε arbitrarily small. Thus for every $n \geq 7$ there is a counterexample on n nodes that is invisible in the $\varepsilon \to 0$ limit.*

Sketch. Lemma 6.1 gives $\Phi = z(z^2 + z + 1)\big((a+7)z(z^2 + z + 1) + 2\big)$ identically in a (non-edge picture); the cubic factor has all roots in $\mathrm{Re}\, z > -1/2$ iff $a + 7 > 16/3$; in the adjacency picture, at $z_0 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ one has $1/\hat{s}'(z_0) = \frac{1}{4} - \frac{\sqrt{3}}{12}i$, so $R^* = 1/4 > 1/8$ for every a; and the class vector at $\beta = 2$ is $(6, 2, 2, 18, 6, 2, 10)/(18a + 17) > 0$. Apply Lemma 6.4. Instances $a = 0, \ldots, 5$ are certified at $t = 20$, $\beta = 821/400$, $(\varepsilon, \delta) = (1/21, 421/8400)$, $q = 421/420$.[24] □

[22] `notes/research_wf3/theory/family_F_certificate.json` (script `family_F_proof.py`); witnesses for $k = 1, \ldots, 6, 8, 10$ ($n = 7, \ldots, 16$) at $(\varepsilon, \delta) = (1/21, 11/210)$ or $(1/21, 11/175)$.

[23] `notes/research_wf3/theory/families_catalogue_n6.json`, `families_catalogue_n7.json`.

[24] `notes/research_wf3/theory/family_E_certificate_part1.json`, `family_E_witnesses.json`.

The two-parameter engine applied to E_1 returns one stable cell, over the same β-cell and with the upper t-end at $\beta = 53/26$ in the same isolating bracket $(53, 54)$ as for E_0.[25] The two regions are not identical: at $\beta = 53/26$, $t = 107/2$ the full support of E_0 is stable and that of E_1 is not (Hurwitz determinant 7 negative), so the upper end of the window depends on a. The cube-root factor arises from the root locus of the class size: for the blow-up family of H_3 one has $\Phi_k(\omega) = (k-1)\omega^2$ at a primitive cube root of unity ω, so the locus passes through ω exactly at class size 2. A float-steered scan of all one-node extensions of the 124 seven-node graphs, followed by exact triage of the 9,808 extensions it flagged (an eigenvalue of $A|_\perp$ near a primitive cube root of unity), found 15 eight-node graphs that satisfy every hypothesis of Lemma 6.4, each with an exact legal witness; extensions not flagged by the float prefilter were not examined exactly.[26]

6.4 Growing thresholds

Theorem 6.6 (Families Q, Q2, Y; proved, refereed). *(Q) For the core* `&F^S{^ReL|?` *with one sink class of size k ($n = 7 + k$), every member is positive at every $\beta > 2$, and is limit-stable at $\beta = 5/2$, 27/10, 14/5 for all $k \geq 25$, 79, 286 respectively. So for every $n \geq 32$ (86, 293) there are counterexamples on n nodes with q arbitrarily close to 3/2 (17/10, 9/5); along the family the limit value of q tends to* $1.84416\ldots$. *(Q2) For the core* `&F^U|m^QTq?` *with two sink classes of sizes k and $3k$ ($n = 7+4k$), limit polynomial $z^3(2z^2+z+2)^2/4$, every member is positive at every $\beta > 2$, and is limit-stable at $\beta = 5/2$, 3, 7/2, 19/5 for all $k \geq 1$, 3, 10, 33. (Y) For the eight-node core* `&G^ZZZr|KjqTo` *with two sink classes of sizes $4k$ and k ($n = 8 + 5k$), every member is positive at every $\beta > 2$, and is limit-stable at $\beta = 3$, 7/2, 4, 9/2, 24/5 for all $k \geq 13$, 24, 46, 117, 363; the family ceiling is $\beta \approx 4.9607$.*

Each threshold is proved sharp by exact real-root isolation in k of the Hurwitz determinants of the shifted limit polynomial. The eight-node core of (Y) is not among the hand-verified graphs.[27] Certified instances include $n = 32$ at $q = 15000/10001$ (Q), $n = 19$ at $q = 20000/10001$ and $n = 47$ at $q = 25000/10001$ (Q2), and $n = 73$ at $q = 20000/10001$ (Y). Exact lower bounds at small n found by search and certified pointwise are listed in Table 3.

n	graph	(ε, δ)	q (exact)
8	`&G^ZnRba\ba^w`	(1/1001, 1709/1001000)	$1709/1001 \approx 1.7073$
9	`&H^\us{\HrEa^|~_`	(1/1001, 17691/10010000)	$17691/10010 \approx 1.7673$
10	`&I^Qz|NbkfuNa^}~|tw`	(1/10001, 9209/50005000)	$18418/10001 \approx 1.8416$
11	`&J^R\~R{\a^k^a^~^~]n^~?`	(1/100001, 861/45455000)	$17220/9091 \approx 1.8942$

Table 3: Certified lower bounds $q_n^* \geq q$ (not hand-verified).

6.5 $q_n^* \to \infty$

A fixed core with sink classes cannot push the limit threshold past the number of core nodes: the limit matrix $D - \bar{b}\mathbf{1}^T$ has trace ≤ -1 when no class is a join. The way around this is a family of cores whose limit polynomial can be prescribed.

Theorem 6.7 (Unbounded thresholds; proved, refereed). *For every integer $N \geq 3$ there is $n_0(N)$ such that for every $n \geq n_0(N)$ some non-complete n-node digraph has a stable positive full-support fixed point at the legal points $(\varepsilon, \delta) = (1/(1+t), (N-1)/(1+t))$ for all large t. Hence $q_n^* \geq N - 1$ for $n \geq n_0(N)$, and $q_n^* \to \infty$.*

[25] `certificates/n8_instances_region2d.json`.

[26] `notes/research_wf3/theory/cube_n8_witnesses.json`.

[27] `notes/research_wf3/theory/family_Q_certificate.json`, `family_Q2_certificate.json`, `family_Y_certificate.json`.

Sketch. Arms forest. In the non-edge digraph D, take a directed 3-cycle $c_0 \to c_1 \to c_2 \to c_0$ and attach "arms" $c_2 \to a \to z_a \to$ (target), so that every node other than c_2 has exactly one D-out-neighbour and reaches c_2 (an in-tree rooted at c_2), while c_2 sends to c_0 and to the first node of each arm. By the matrix determinant lemma (Lemma T of the notes), $\chi_{D-e_{c_2}\mathbf{1}^T}(z) = z^{m-d}p(z)$ for any $p = z^d + z^{d-1} + c_2 z^{d-2} + \cdots + c_d$ with integers $c_2 \geq 1$, $c_j \geq 0$ satisfying a mild forest constraint; the core has $m = 2p(1) - 3$ nodes. Lemma Q (Sherman–Morrison) gives ψ_D in closed form and Lemma P gives positivity in closed form. A single sink class receiving from every core node except c_2 has $\Phi_k = z^{m-d}(G + kp)$ with G fixed. *Target polynomial.* With $s = 1/(2N)$ and an integer W, round the coefficients of $\prod_{k=1}^{N}\big((z+s)^2 + W^2k^2\big)$ (except the first two) to the nearest integer. By Rouché (Lemma S), for $W > W_0(N) = 2N\,4^N(N+1)^{2N-2}$ all roots of the rounded polynomial $\tilde{p}$ satisfy $-3/(4N) < \operatorname{Re} z < -1/(4N)$, and all its coefficients from c_2 on are ≥ 1. *Positivity* at $\beta = N$ holds for large W because the top coefficient dominates. *Class size.* $\Phi_k/(m+k) \to \chi_L = z^{m-d}\tilde{p}$, so by root continuity there is k_0 with all roots of Φ_k in $\operatorname{Re} z > -1/N$ for $k \geq k_0$. Lemma 2.4 at the rational $\beta = N$ finishes, with $n_0 = m + k_0$.[28] □

The bound $n_0(N)$ is super-exponential, and k_0 is not explicit in general. Exact reduced certificates give $n_0 =$ 1,812, 6,214,995 and 1,590,755,528 for $N = 3, 4, 5$; a 65-node core with p-coefficients $(1, 1, 11, 7, 10, 3, 1)$ gives $q_n^* \geq 3$, 4, 9/2, 23/5 for all $n \geq 219$, 657, 2,475, 4,909. Two explicit graphs are certified directly by the pointwise verifier without using any lemma: $n = 28$ at $q = 20000/10001$ and $n = 41$ at $q = 22500/10001$; at $k_0 - 1$ the explicit limit test fails with a negative Hurwitz determinant, so the reduced thresholds are sharp there.[29] The only known upper bound remains $q_n^* \lesssim n-2$ from (2).

6.6 Unbounded support size at one parameter point

Theorem 6.3 uses a k-dependent schedule $\beta \to 2$. At a fixed legal point the family F_k eventually dies: at $(\varepsilon, \delta) = (1/21, 11/210)$ it is stable exactly for $k = 0, \ldots, 40$ and unstable for every $k \geq 41$, and in fact at every fixed legal point for large k, because χ_L has the factor $z^2 + z + 1$ with roots on $\operatorname{Re} z = -1/2$. A different core avoids this.

Let $R =$ `&F^U|m^QTq?` (one of the 124; rows `0111110 1011011 1101101 1100111 1101001 0010101 1100100`), $S = (1, 1, 0, 1, 1, 1, 1)$ and $R_k := R[S, \text{all}; k]$, $n = 7 + k$.

Theorem 6.8 (One point, every size; proved, refereed)**.** *At $(\varepsilon, \delta) = (1/21, 11/210)$, i.e. $t = 20$, $\beta = 21/10$, $q = 22/21$, the full support of R_k is a stable positive non-clique fixed point for every $k \geq 0$. Consequently, at this point a non-complete n-node digraph with a stable positive full-support fixed point exists if and only if $n \geq 6$.*

Sketch. In the non-edge picture, (3) holds iff every eigenvalue of $D' + aJ$, $a = t/\beta$, has real part $> -1/\beta$, and for sink-class families $\det(zI - aJ - D') = z^{p-1}\Psi(z)$ with $\Psi = z\det(zI - D - aJ) - a\sum_j k_j \det(zI - D + b_j\mathbf{1}^T)$. Here $\Psi_k = z^3 g_k$ with $g_k(z) = z^5 - 3z^2 - 2z - 1 - a[(k+7)z^4 + (k+12)z^3 + (3k+21)z^2 + (k+13)z + (k+5)]$. At $a = 200/21$, after the shift $z = -10/21 - x$, every Hurwitz determinant is a polynomial in k with only positive coefficients (for example $21\Delta_1 = 200k + 1450$), hence positive for all $k \geq 0$. Positivity: $w_R(21/10) > 0$ and $\sigma_S = 381654/295819 > 1$, so Lemma 6.2(X2) applies for every k. The "only if" part is Theorem 4.1; $n = 6$ is G_1.[30] □

An independent second proof (equitable-partition quotient, separate exact code) confirms the result; members up to $n = 107$ were certified pointwise. Also proved in the notes (the referee spot-checked the first statement at 28 exact legal points and re-proved the self-join statement independently): R_k

[28] Theorem U, `notes/theorem_U/`; referee record `notes/theorem_U_workflow.json`.

[29] `certificates/theorem_U_explicit_instances.json`.

[30] Theorem A, `notes/theorem_A_fixed/`; `certificates/theorem_A_fixed_parameter_instances.json` ($k = 0, \ldots, 15$); referee record `notes/theorem_A_fixed_workflow.json`.

is an SFP at *every* legal point with $\beta \le 43/20$, for every $k \ge 0$; every legal point with $\beta < 4.9607$ carries an infinite family (via Theorem 6.6(Y)); and, conditionally on the lemmas of Theorem 6.7, every legal point does, with an exact instance at $(\varepsilon, \delta) = (1/2, 2)$ for all $k \ge 1581$. The k-fold self-join of R ($n = 7k$), which is not a blow-up, is an SFP at every legal point with $\beta \le 21/10$ for every $k \ge 1$.

Proposition 6.9 (A refuted conjecture; certified)**.** *It is not true that positivity of* $(D^T + rI)^{-1}\mathbf{1}$ *for some* $r \in (0, 1/2)$ *forces* D *to have an eigenvalue with real part* $\le -1/2$*. For example, the non-edge matrix of* `&F^U|m^QTq?` *has every eigenvalue with real part* $> -10/21$ *while* $(D^T + \frac{10}{21}I)^{-1}\mathbf{1} > 0$*; and* 33 *of the* 54 *catalogue entries with* S *equal to all nodes, e.g. the base* `&FMAlW\d\y?` *with* $\chi_D = z^7 - z^5 - 7z^4 - 10z^3 - 9z^2 - 4z - 1$*, have every eigenvalue of* D *in* $\mathrm{Re}\, z > -1/2$ *and a positive vector at* $r = 49/100$*. In particular the joins* $G \vee K_k$ *of such bases are limit-stable for every* $k \ge 1$*.*

(This was "Conjecture P" of the project notes; see Question 8.4.)

7 Verification and reproducibility

7.1 Certificates

Every number in Sections 4–6 is read from a certificate file that can be regenerated by a named script (Appendix A). Values are stored exactly: integers, rationals, and algebraic numbers as an irreducible integer minimal polynomial together with a rational isolating interval; decimals appear only as labelled approximations. A region certificate stores, for each cell, the isolating data of its walls, an exact sample, the verdict with its reason (for example "Hurwitz determinant 6 negative"), and the outcome of the pointwise cross-check. The "never" records of the large sweeps store critical values and cell counts rather than full decompositions, so verifying them means re-running the committed engine; the records of all stable graphs are self-contained. Bulk files carry SHA-256 checksums.

7.2 Independent audits

The engine and the $n = 6$ map were audited adversarially, including a from-scratch symbolic re-derivation of the region of H_3 (same set and same minimal polynomial). The $n = 7$ prefilter and resolver were audited twice (979,800 exact comparisons; all 786,596 undecided lines recomputed independently, byte-identical), and the ray reduction, curve test and edge bound once (an independent all-minors computation on 3,129 graphs and 6,000 random exact legal points on never-stable graphs). The family theorems (Theorems 6.3, 6.5, 6.6, 6.7, 6.8), the 626-entry catalogue and the threshold tool were each refereed by a separate reviewer who re-derived the steps, taking Lemma 2.4(b),(c) as given; no blocking issue was found. Non-fatal defects found by the referees (e.g. a misprinted node count, $2p(1) - 1$ for $2p(1) - 3$, in the notes on Theorem 6.7) are logged and corrected here.

7.3 Human verification

The two new six-node counterexamples were hand-verified on 14 October 2026, and all 124 seven-node counterexamples on 15 October 2026; the project owner recorded the latter hand check as not fully confident and made sign-off conditional on the machine re-verification below. The 124 were additionally re-decided by four exact verifiers written from the definitions without access to the project's code: one with exact rationals and its own Hurwitz determinants; one with pure fractions and its own elimination and characteristic polynomials (with the transpose convention reversed, 0 of the 124 survive, a useful convention check); one by exact right-half-plane root counting without Hurwitz determinants (Routh array, Sturm sequences and exact isolation, required to agree); and one checking structural claims (pairwise non-isomorphism, domination-freeness, non-completeness, the exact limit test and

$\alpha(E_0) = -1/2$). All 124 were confirmed at 742 legal points with 0 undecided cases.[31] The verification also found that 15 of the first stored witnesses in the limit file are illegal points ($q \le 1$); a separate file lists one legal witness per graph. Graphs with $n \ge 8$ nodes have not been hand-verified.

7.4 Nondegeneracy

The conjecture is usually stated for nondegenerate CTLNs. At a witness point of each of the three six-node and 124 seven-node counterexamples we certified exactly that every one of the $2^n - 1$ supports has $\det(I - W_\sigma) \neq 0$, a fixed point with no zero coordinate and no off-support drive equal to 0, and that the full support is stable there.[32]

7.5 Machine-checked statements

Lean 4 (toolchain v4.35.0) with Mathlib (tag v4.35.0) checks 87 theorems with no `sorry`; the axiom audit shows only `propext`, `Classical.choice` and `Quot.sound`. They comprise: the closed form (4) with the transpose and $I - W = \varepsilon K$; the legality equivalence (5); Hurwitz stability of $-I + W$ iff positive stability of K; the ray identity, characteristic-polynomial scaling and propagation of stability down rays (Lemma 3.3); the intermediate-value step with the explicit root, and the forward direction of Lemma 3.4; Lemma 2.3; Lemma 3.6 including its spectral step; Lemma 3.5 with its induced-subgraph form; Stodola's condition (Lemma 3.7); and the twin eigenvalue (Lemma 3.8). Not formalised: the converse of Lemma 3.4 (it needs continuity of eigenvalues), the limit criterion Lemma 2.4, the rank-one identities of Lemma 2.2(ii)–(v), the off-support condition, and the full certificate lemma "(1)–(3) imply a stable equilibrium of (1) with support σ".[33]

Trust base. The certificates rely on: exact polynomial arithmetic of a standard computer-algebra system for the projection polynomials of the decomposition (determinants, factorisation, gcd, resultants, discriminants; the projection identities were re-verified by independent fraction-free determinants on a sample of 321 records); the Routh–Hurwitz theorem, which is cited; nauty's `geng`/`directg` for the seven-node enumeration, validated by its count; and the C prefilters, validated by reproducing the six-node data exactly.

8 Open problems

Question 8.1 (Growth rate)**.** How fast does q_n^* grow? Theorem 6.7 gives $q_n^* \to \infty$ with a super-exponential $n_0(N)$; the certified points are q close to 2 at $n = 19$, 5/2 at $n = 47$ and 23/5 for $n \ge 4{,}909$, while the only upper bound is Geneson's $n - 2$. Is $q_n^* \ge c \log n$, or $q_n^* \ge n^c$? The bottleneck is integer polynomials with $c_1 = 1$, nonnegative coefficients, roots in a thin strip $\operatorname{Re} z > -1/N$, and small value at 1.

Question 8.2 (The $n = 8$ map)**.** Complete the $\varepsilon \to 0$ classification at $n = 8$. An overflow-guarded spectral prefilter reproduces all $n \le 7$ data exactly; the full pass is estimated at 500–800 CPU-hours, and besides the family members of Section 6 one further eight-node limit-stable graph, `&GLjlNZ}NzH\o` (41 edges; certified, not hand-verified), was found in a random dense sample. The finite-ε map at $n = 8$ (1.79 trillion digraphs) is out of reach by enumeration alone.

Question 8.3 (Formalisation)**.** Formalise the converse of Lemma 3.4 (openness of positive stability), the limit criterion Lemma 2.4, and the certificate lemma (1)–(3)$\Rightarrow$ stable equilibrium, so that every certified case becomes a checked instance.

[31] `notes/verify_124_workflow.json`; `certificates/n7_candidates_legal_witnesses.json`.

[32] `certificates/nondegeneracy_witnesses.json`, script `n7/nondegeneracy_check.py`.

[33] `notes/lean/README.md`, `notes/lean/CtlnCert/`.

Question 8.4 (Characterising limit-stable graphs)**.** Proposition 6.9 refutes the natural spectral obstruction. Which graphs are limit-stable, i.e. have $w(\gamma) > 0$ and $\alpha < -\gamma$ at a common $\beta > 2$? A related question: is $\alpha \leq -1/2$ for every graph whose full support is an SFP at some legal point? This holds for $n \leq 7$ (Corollary 5.2), with equality attained for every $n \geq 7$ by Theorem 6.5; no proof is known.

Question 8.5 (Proper supports)**.** The truth of the conjecture depends only on full supports (Theorem 2.6), but listing all pairs (G, σ) realising a counterexample needs condition (2): describe the admissible in-sets for the $n \leq 7$ counterexamples. At $n = 7$ all 12,288 pairs formed by a six-node counterexample and a seventh node with any in/out pattern have been certified (7,744 stable somewhere, 4,544 never, 0 escalations; `certificates/n7_proper_supports_all.jsonl`; not hand-verified); a structural description is open.

A Certificates

Paths are relative to the project archive. "Script" is the program that regenerates the file.

claim	certificate	script
Geneson's graph reproduced	`certificates/stage0_geneson_2607.21396.json`	test suite, block 8
$n = 5$ package, diff zero	`certificates/stage1_n5_diff.json`	`stage1_n5_diff.py`
engine validation	`certificates/validate_fastregion_n4.json`, `_n5.json`, `region2d_v2_validation.json`	`validate_fastregion.py`, `region2d.py`
$k \leq 4$	`certificates/region2d_n2to4_all.json`	`region2d.py`
Theorem 4.1	`certificates/region2d_n5_shard0of1*`, `n5_summary.json`	`region2d.py`
Theorem 4.2	`certificates/n6_summary.json`, `region_n6_limit_*`, `region2d_n6_shard*`, `region_n6_eps*`, `COUNTEREXAMPLE_CANDIDATE_n6_idx*.json`	`fastregion.py`, `region2d.py`
twin classes	`certificates/n6_survivor_equivalence_classes.json`	`notes/prover_scripts_J/`
Theorem 4.4	`certificates/qstar/qstar_n6.json`	`n7/qstar.py`
Table 2	`certificates/n7_finite_summary.json`, `n7_curve_TOTAL.json`, `n7_region2d/`	`n7/run_n7_curve.py`, `run_n7_region2d.py`, `aggregate_n7.py`
limit map $n = 7$	`certificates/COUNTEREXAMPLE_CANDIDATES_n7_limit.jsonl`, `n7_limit_tally.json`	`n7/run_n7_limit.py`
the 124 graphs	`certificates/COUNTEREXAMPLE_CANDIDATES_n7_finite.jsonl`, `n7_candidates_legal_witnesses.json`	`n7/aggregate_n7.py`, `notes/verify_124/`
E_0	`certificates/COUNTEREXAMPLE_CANDIDATE_n7_finite_FNFL.json`, `qstar/qstar_fnfl.json`	`n7/write_finite_certificates.py`, `n7/qstar.py`
Proposition 5.3	`notes/lemma_M_refuted.md`, §2	`notes/prover_scripts_M/`
Proposition 5.5	`certificates/n7_bulge_Rstar_exact.json`, `n7_limit_hits_finite_eps_bulge_witnesses.json`	`notes/bulge_exact/`
Theorem 5.6	`certificates/qstar/qstar_n7_finite.json`, `qstar_n7limit.json`	`n7/qstar.py--sharpen`
Theorems 6.3, 6.5, 6.6, Table 3	`notes/research_wf3/theory/family_*_certificate.json`, `family_E_witnesses.json`, `top_n8.json`, `cert_q_910.json`, `cert_q_11.json`, `families_catalogue_n{6,7}.json`	`family_*_proof.py`, `certify_top.py`, `certify_q.py`, `run_family_prover.py`, `verify_all.py`
eight-node instances	`certificates/n8_instances_region2d.json`, `notes/research_wf3/theory/cube_n8_witnesses.json`	`n7/n8_instances_region2d.py`, `witness_cube.py`
Theorem 6.7	`certificates/theorem_U_explicit_instances.json`, `notes/theorem_U/`	`theorem_U.py`, `certify_forest.py`, `certify_explicit.py`

claim	certificate	script
Theorem 6.8	`certificates/theorem_A_fixed_parameter_instances.json`, `notes/theorem_A_fixed/`	`theorem_A.py`, `independent_A.py`
nondegeneracy	`certificates/nondegeneracy_witnesses.json`	`n7/nondegeneracy_check.py`
Lean	`notes/lean/CtlnCert/`	`lake build`, `Audit.lean`

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